Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/18/2016
PI Name:	Lindner, Jonathan R M.D.		
Project Title:	Biomarker Assessment for Identifying Heightened Risk for Cardiovascular Complications During Long-duration Space Missions		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lindnerj@ohsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	503-494-0236
Organization Name:	Oregon Health & Science University		
PI Address 1:	3181 SW Sam Jackson Park Road		
PI Address 2:	Cardiology UHN62		
PI Web Page:			
City:	Portland	**State:**	OR
Zip Code:	97239-3011	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	06/01/2015	**End Date:**	05/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: change in period of performance to 6/1/2015-5/31/2017 (originally 5/1/15-4/30/17), per NSBRI (Ed., 6/29/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fazio, Sergio M.D., Ph.D. (Oregon Health & Science University) Spellman, Paul Ph.D. (Oregon Health & Science University) Stevens, Jan (Oregon Health & Science University) Raber, Jacob Ph.D. (Oregon Health & Science University) Turker, Mitchell Ph.D. (Oregon Health & Science University) Ferencik, Maros M.D. (Oregon Health & Science University)		
Grant/Contract No.:	NCC 9-58-CA04201		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goals for atherosclerotic screening of astronauts in preparation for space exploration are quite different than that used for screening of the general population. Since an in-flight atherosclerotic complications would put the individual and the entire mission at great peril, any screening algorithm is required to definitively identify the predisposition to future atherothrombic events irrespective of the presence of significant stenosis. The aim of this proposal is to use state-of-the-art omics approaches for improving atherosclerotic risk assessment in apparently healthy individuals. Our proposal is divided into two main aims; first to identify representative individuals with various risk for cardiovascular outcomes and second to use personal metabolomics and lipidomics profiling to correlate risk of disease with molecular biomarkers. Risk will be stratified in a two-step process, first involving the recruitment of asymptomatic subjects with either very low, moderate, or very high risk features on coronary CT angiography features. A second evaluation method (myocardial contrast echocardiography) will then be applied to assess endothelial function within each cohort since the vascular endothelium is a key target in the pathophysiology of accelerated atherosclerotic disease during deep space exploration. For each risk category group, we will perform integrative omic analysis from serum and peripheral blood monocytes, which will focus on differences in unbiased metabolomics, lipidomic, and oxidative proteomics using mass spectroscopy and ultra performance liquid chromatography (UPLC). In addition, whole genomic sequencing will be performed on peripheral blood mononuclear cells (PBMCs) from as many as 20 individuals for use in a disease risk evaluation. Any omics patterns that are identified as a strong predictor of risk may be particularly useful for identifying individuals with heightened risk for the effects of prolonged space exploration. These tests may not only be useful for initial screening, but also for periodic evaluation during mission lead time and also for investigational assessment of the effects of space radiation on the International Space Station (ISS) or during space exploration missions. To perform these studies, we have assembled a multidisciplinary team of investigators with expertise in non-invasive cardiovascular imaging in risk assessment, metabolomics, proteomics, endothelial biology, and bioinformatics/biostatistics. Key findings: Imaging analysis has been completed on a total of 15 of the enrolled subjects. We have been able to obtain robust stress-rest myocardial perfusion imaging data with myocardial contrast echocardiography (MCE) which has informed us that the technology is performing as expected. We are encouraged that about half of the "normal CT angiography" cohort have been classified by MCE as having microvascular dysfunction whereas the other half have normal microvascular function. We believe that this rather high rate of microvascular dysfunction is reflective of referral bias where a high proportion of those sent for CT-A but without epicardial coronary artery disease have symptoms of angina from other causes such as mirovascular dysfunction. Blood has been received and is being processed for metababolomic, lipidomic, and whole gene sequencing analysis. For the coming year, we will continue to enroll and study patients. Because of the nature of the analysis (machine-based learning) the primary analysis of evaluating whether omics patterns are predictive of risk based on the dual stratification with macro- and microvascular imaging will not be possible until all omics data are completed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. We have started to understand the prevalence of microvascular dysfunction in those referred for CT-A but with negative results. 2. Any "omics" patterns of enhanced risk may provide the basis for evaluating new risk assessment strategies both for CAD and for vascular complications of radiation therapy. 3. "omics' patterns for microvascular dysfunction have never been assessed but will provide critical information on pathobiology.
Task Progress:	1. The study was initiated in fall 2015. 2. We have completed the study protocol in 15 individuals. 3. Subject recruitment and study will continue. 4. Interim analysis is not possible due to the statistical methods that are being used to integrate "omics" and imaging stratification data.
Bibliography Type:	Description: (Last Updated: 05/26/2021)
Significant Media Coverage	Terry L. "OHSU-led study could help NASA’s mission to Mars." Newspaper article about PI's research in The Oregonian, June 26, 2015. http://www.oregonlive.com/health/index.ssf/2015/06/ohsu_osu_study_could_help_nasa.html ; accessed 5/18/16., Jun-2015